and Beans. Those seeds with endosperms are called endospermous or albuminous seeds. The endosperms in these seeds supply nutrition to the embryo during seed germination. Examples: Paddy, Coconut and Castor.



Aleurone tissue consists of highly specialised cells of one or few layers which are found around the endosperm of cereals (barley and maize). Aleurone grain contains sphaerosomes. During seed germination cells secrete certain hydrolytic enzymes like amylases, proteases which digest reserved food material present in the endosperm cells.

Ruminate endosperm: The endosperm with irregularity and unevenness in its surface forms ruminate endosperm. Examples :*Areca catechu, Passiflora* and *Myristica*

Functions of endosperm:

- It is the nutritive tissue for the developing embryo.
- In majority of angiosperms, the zygote divides only after the development of endosperm.
- Endosperm regulates the precise mode of embryo development.



Coconut milk is a basic nutrient medium which induces the differentiation of embryo (embryoids) and plantlets from various plant tissues. Coconut water from tender coconut is free-nuclear endosperm and white kernel part is cellular.

Development of Dicot embryo

The Stages involved in the development of Dicot embryo (*Capsella bursa-pastoris* – Onagrad or crucifer type) is given in Figure 1.22. The embryo develops at micropylar end of embryo sac. The zygote undergoes transverse division to form upper or terminal cell and lower or basal cell. Further divisions in the zygote during the development lead to the formation of embryo. Embryo undergoes globular, heart shaped stages before reaching a mature stage. The mature embryo has a **radicle,** two **cotyledons** and a **plumule**.

Activity

Collect the fruits of *Tridax* (Cypsella). Using a needle dissect out the content, separate the embryo and observe different stages of dicot embryo – globular, torpedo, heart shaped under a dissection microscope.

Seed

The fertilized ovule is called seed and possesses an embryo, endosperm and a protective coat. Seeds may be endospermous (wheat, maize, barley and sunflower) or non endospermous. (Bean, Mango, Orchids and cucurbits).



Fresh weight of an orchid seed may be 20.33 microgram and that of double coconut (*Lodoicea maldivica*) is about 6 kg.

***Cicer* seed (example for Dicot seed)**

The mature seeds are attached to the fruit wall by a stalk called **funiculus**. The funiculus disappears leaving a scar called **hilum**. Below the hilum a small pore called **micropyle** is present. It facilitates entry of oxygen and water into the seeds during germination. Each seed has a thick outer covering called seed coat. The seed coat is developed from integuments of the ovule. The outer coat is called **testa** and is hard whereas the inner coat is thin, membranous and is called **tegmen**. In Pea plant the tegmen and testa are fused. Two cotyledons laterally attached to the embryonic axis and store the food materials in pea whereas in other seeds like castor the

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